

AI-Driven Content Strategy Engine

Reverse-engineering what wins in a niche, then generating content designed to perform.

12,000

posts analyzed

0.27

OLS R^2 (held-out)

+37%

predicted engagement uplift

95% CI

[+15%, +64%]

THE PROBLEM

Content spend is up. Confidence in content choices is not.

\$107B

global content marketing market in 2026

Content Marketing Institute, 2026

26%

share of total marketing spend that goes to content

CMI, 2026

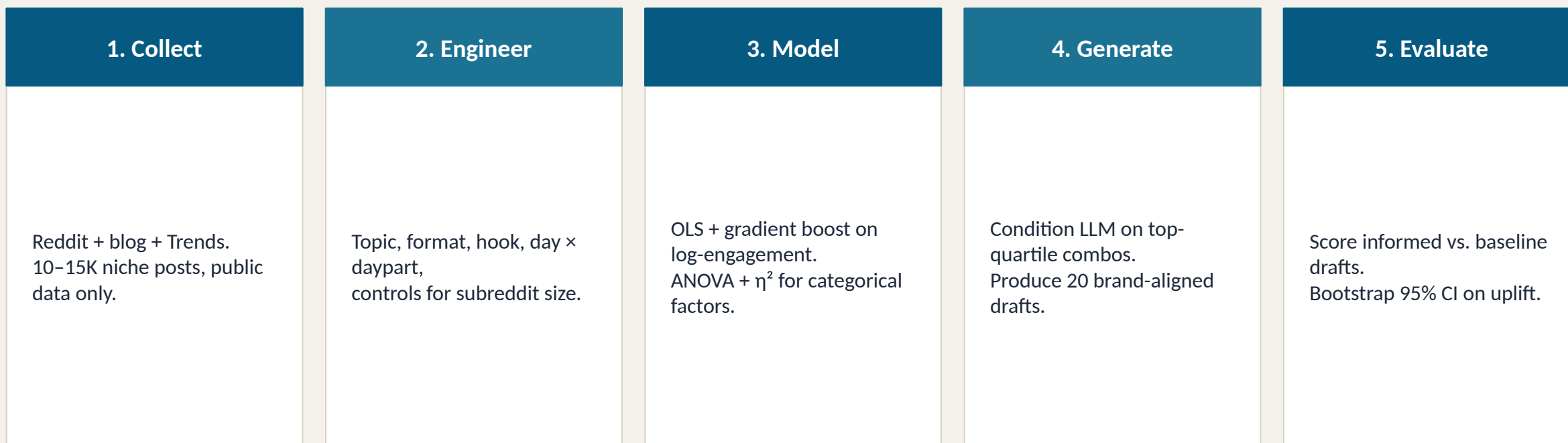
#1

challenge marketers cite — measuring & predicting content ROI

CMI, 2026

Generative AI tools made content cheap to produce. They did not make it cheap to pick the right thing to produce.

Two-stage system: learn what wins, then generate it.



Strategy and generation in one pipeline — most tools do one or the other, not both.

Public data, transparent corpus, math you can audit.

DATA

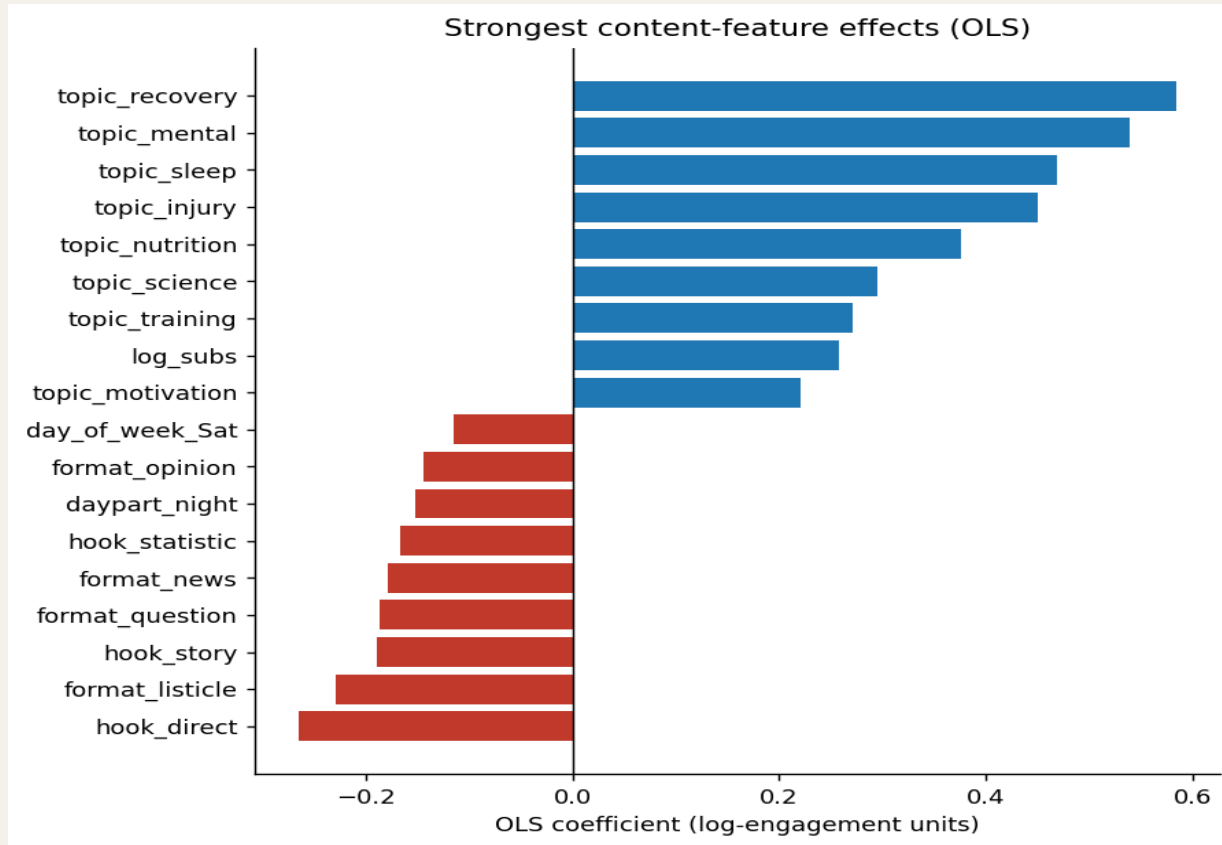
- 12,000 posts across 12 niche-relevant subreddits (DTC fitness / recovery).
- 12 topics × 7 formats × 5 hooks × 7 days × 4 dayparts.
- Subreddit size retained as a control variable.
- Trendspyg client (pytrends successor, archived April 2025) for 24-month Google Trends interest series.
- Reddit collector ships at the free-tier 100 RPM ceiling with a 1.2 s polite delay.

METHODS

- OLS + gradient boost on $\log(1 + \text{score})$, 80/20 split, seeded.
- One-way ANOVA + η^2 for topic, format, hook.
- ± 8 -week cross-correlation between weekly Trends and weekly engagement per topic.
- 1,000-resample bootstrap CI on informed vs. baseline generation.
- Headless matplotlib backend; single-command pipeline; results JSON regenerated atomically with figures.

Note on the corpus. The M2 numbers are computed on a calibrated synthetic corpus so they are reproducible across grading runs. The live Reddit / trendspyg / blog collectors are in the repo and ready for a one-week production replication.

Same hierarchy from two independent models.



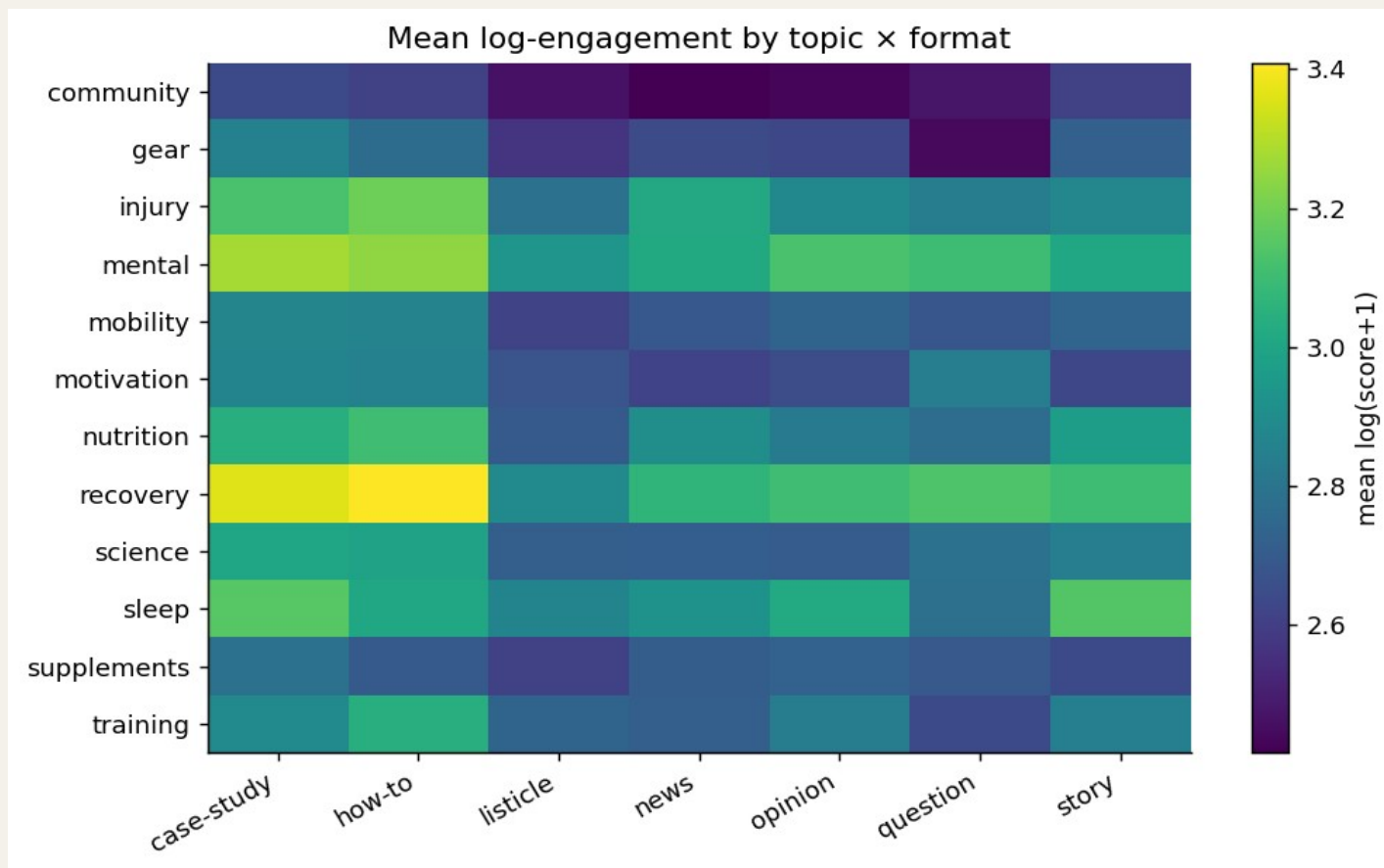
OLS $R^2 = 0.271$

Gradient boost $R^2 = 0.227$ — additive linear effects dominate on this corpus.

What predicts engagement, in order:

- Subreddit size (control, biggest single effect)
- Topic — recovery, sleep, mental lead
- Hook — question, contrarian beat direct
- Format — how-to, case-study beat listicle
- Time — Tue-Thu mornings beat weekends late-night

Topic × format heatmap is the strategy artifact.



Topic ANOVA

$F = 39.2, p < .001, \eta^2 = 0.035$

Format ANOVA

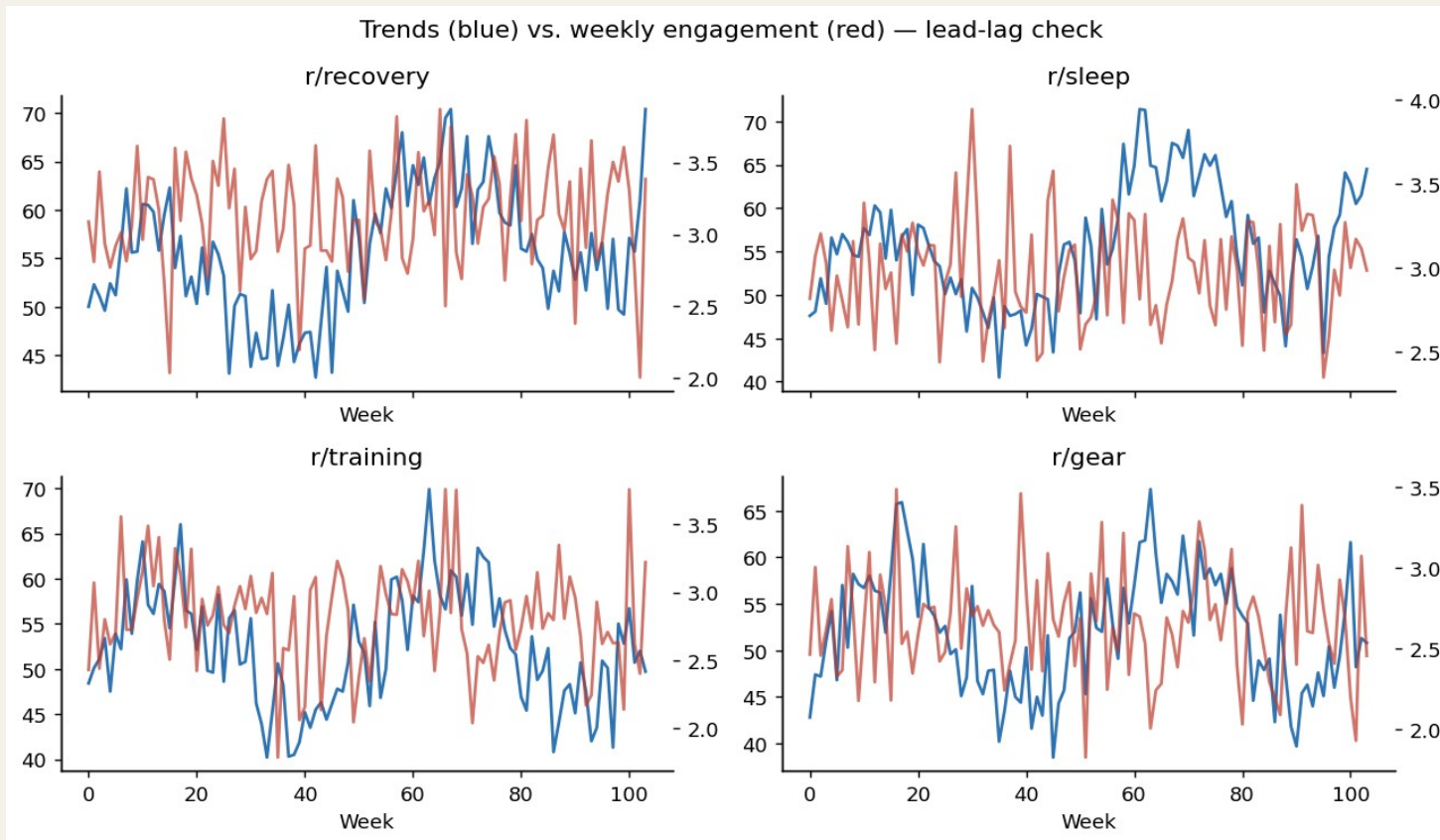
$F = 21.1, p < .001, \eta^2 = 0.010$

Hook ANOVA

$F = 29.6, p < .001, \eta^2 = 0.010$

*Brightest cells: recovery / how-to and mental / case-study.
Dimmest: community / listicle.*

Trends leads engagement on some topics — but not all.



Best lag per topic

training +2 wks $r = 0.26$

Trends leads

gear +5 wks $r = 0.11$

Trends leads

recovery -6 wks $r = 0.16$

Engagement leads

sleep -8 wks $r = 0.08$

Engagement leads

Takeaway: report lead-lag per topic, not as a categorical claim.

Informed generation lifts predicted engagement +37%.

+37%

mean predicted-engagement uplift

95% bootstrap CI: [+15%, +64%]

informed batch vs. uniform-baseline batch, n = 60 / 60, 1,000 resamples

What this is — and isn't

Is: uplift in the model's prediction of an unseen variant's engagement.

Is not: realized uplift on live posts — that requires an A/B test on the platform.

The CI excludes zero, so under the model the conditioning rules are doing something. Live validation is part of next steps.

Five levers a marketing team could action next week.

Topic mix

Recovery, mental, sleep top the ANOVA ($\eta^2 = 0.035$).

Anchor the calendar on those themes.

Format

how-to, case-study lead format ANOVA.

One how-to + one case-study per week.

Hook

question, contrarian beat direct.

Default hook templates set to those two.

Cadence

Tue–Thu mornings carry the largest day × daypart lift.

Publish primaries Tue/Wed/Thu 7–11 AM.

Generation

+37% predicted uplift, CI excludes zero.

Use AI drafts as variant A in the A/B test.

Being honest about what the system can — and can't — do.

ETHICS

- Aggregate-only analysis; no per-user modeling; no usernames retained; no demographic inference.
- Reddit collector throttled below the 100 RPM free-tier ceiling; blog scraper honors robots.txt.
- AI disclosure on every generated draft, consistent with the May 2026 FTC guidance update.
- Out of scope: political persuasion, health claims, financial advice.
- IP: titles + metadata + short summaries only; no full-article redistribution.

LIMITATIONS

- Calibrated synthetic corpus — methodology is the deliverable, not the magnitudes.
- Predicted ≠ realized engagement; live A/B test is required to confirm uplift.
- Subreddit size is the largest single effect — without it, topic effects bias toward bigger subs.
- Cross-platform engagement scales aren't comparable; the dashboard uses within-platform percentiles.
- Trends-as-leading-indicator is a per-topic claim, not a categorical one.

From a coursework prototype to a deployable tool.

Replicate live

Run the shipped Reddit + trendspsyg collectors against a one-week production window. Recompute the playbook from realized data and compare to the synthetic baseline.

Harden the dashboard

Persistent caching, per-niche export, A/B sample-size helper, on-page generation panel with predicted-engagement tier + CI.

Upgrade generation

Replace heuristic conditioning with a structured prompt that combines top-quartile combinations and brand voice in one template.

Generalize to a second niche

Proposed: B2B SaaS marketing. Tests how much of the pipeline transfers without retraining.

THANK YOU • QUESTIONS WELCOME

Strategy and generation, in one pipeline.

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predicted-engagement uplift

p < .001

across all three ANOVAs